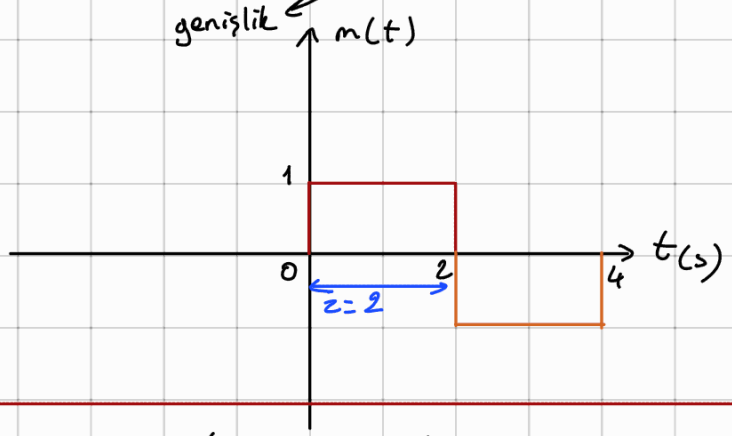
FM Modulation and Demodulation

The message signal $m(t) = \prod\left(\frac{t-1}{2}\right) - \prod(\frac{t-3}{2})$, t in seconds, modulates the carrier signal $c(t) = 5Cos(500\pi t)$ to obtain FM signal $y(t) = 5Cos(500\pi t + \emptyset(t))$, where $\emptyset(t) = 2\pi k_f \int m(\tau)d\tau$ with $k_f = 50$. The demodulator consists of a differentiator, a diode (assume ideal), a low pass filter (LPF) and a capacitor for DC blocking, successively.

1) Plot the message signal. Obtain and plot the magnitude spectrum of the message signal.
2) Obtain and plot the time-domain representation and magnitude spectrum of the phase $\emptyset(t)$.
3) Obtain and plot the time-domain representation and magnitude spectrum of the modulated signal $y(t)$.
4) Obtain and plot the time-domain representation and magnitude spectrum of the signal at each step of the demodulator. (Do the filtering in frequency domain.) Comment on the signal at each step. Compare the message signal and demodulated signal, and comment on them.
5) Do the steps above that can be done analytically, and compare to the simulation results and comment on them.

1) $m(t) = \prod\left(\frac{t-1}{2}\right) - \prod\left(\frac{t-3}{2}\right)$



$x(t) = A\,\Pi(t/z) \Rightarrow X(f) = Az\,\text{sinc}(fz)$

$\Pi\left(\frac{t}{2}\right) \overset{F}{\leftrightarrow} 2\,\text{sinc}(2f)$

Zamanda öteleme özelliği (Timeshifting)

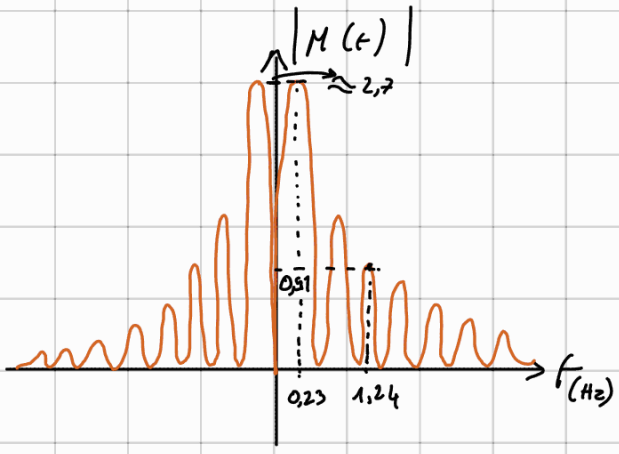
$x(t) \overset{F}{\leftrightarrow} X(f)$

$x(t-t_0) \overset{F}{\leftrightarrow} X(f)e^{-j2\pi f t_0}$

$\Pi\left(\frac{t-1}{2}\right) \overset{F}{\leftrightarrow} 2\,\text{sinc}(2f)e^{-j2\pi f}$

$-\Pi\left(\frac{t-3}{2}\right) \overset{F}{\leftrightarrow} -2\,\text{sinc}(2f)e^{-j2\pi 3f}$

$m(t) \overset{F}{\leftrightarrow} M(f) = 2\,\text{sinc}(2f)\left(e^{-j2\pi f} - e^{-j2\pi 3f}\right)$



$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$

$e^{jx} = \cos x + j\sin x$

$\hookrightarrow M(f) = 2\,\text{sinc}(2f)\left[\cos(\pi f) - j\sin(\pi f) - \cos(3\pi f) + j\sin(3\pi f)\right]$

$\Rightarrow |M(f)| = \sqrt{\left[\cos(\pi f) - \cos(3\pi f)\right]^2 + \left[-\sin(\pi f) + \sin(3\pi f)\right]^2}\ \ |2\,\text{sinc}(2f)|$

so $|M(0)| = 0$, $|M(1{,}24)| \approx 0{,}51$

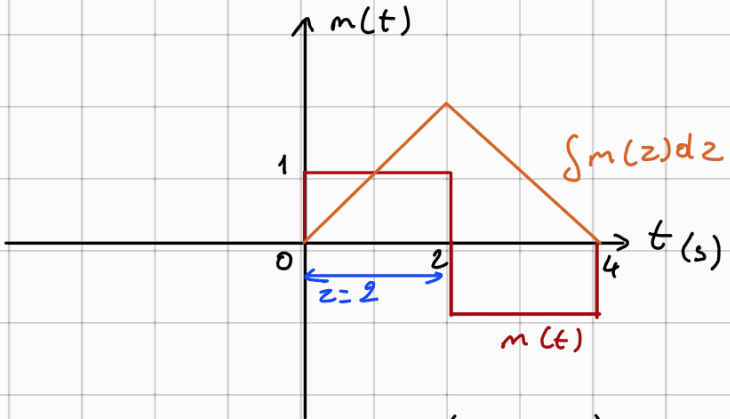
b)
$$\int_{-\infty}^{t} m(z)\,dz = \begin{cases} \int_{-\infty}^{t} 1\,dz = \boxed{t} & t \le 2 \\ \int_{-\infty}^{2} 1\,dz + \int_{2}^{t} -1\,dz \\ 2 - t + 2 = \boxed{4-t} & 2 < t \le 4 \end{cases}$$

$$\phi(t) = 2\pi k_f \int m(z)\,dz$$

when $k_f = 50$

$$\phi(t) = 100\pi \int m(z)\,dz$$

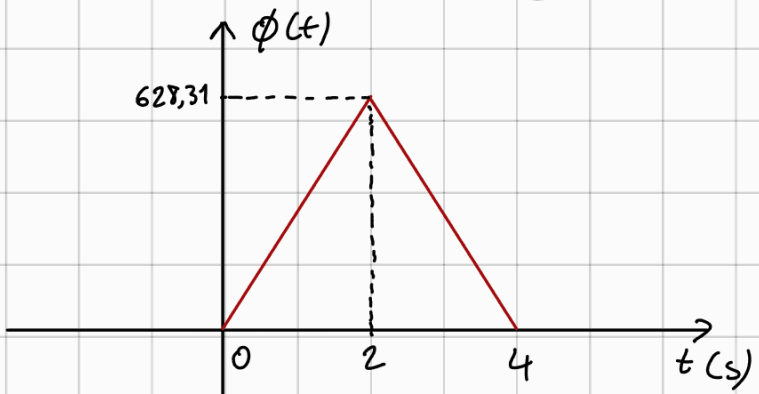
$$m(t) = \Pi\left(\frac{t-1}{2}\right) - \Pi\left(\frac{t-3}{2}\right)$$



$$\Rightarrow \int m(z)\,dz = 2\cdot\Lambda\left(\frac{t-2}{2}\right)$$

$2z = 4 \quad z = 2$

$$\Rightarrow \phi(t) = 100\pi \cdot 2\,\Lambda\left(\frac{t-2}{2}\right)$$
$$= 200\pi\,\Lambda\left(\frac{t-2}{2}\right)$$
$$\approx 628{,}31\,\Lambda\left(\frac{t-2}{2}\right)$$



$$A\,\Lambda\left(\frac{t}{2}\right) \leftrightarrow A2\,\text{sinc}^2(f2)$$

Zamanda öteleme özelliği (Timeshifting)

$x(t) \xleftrightarrow{F} X(f)$

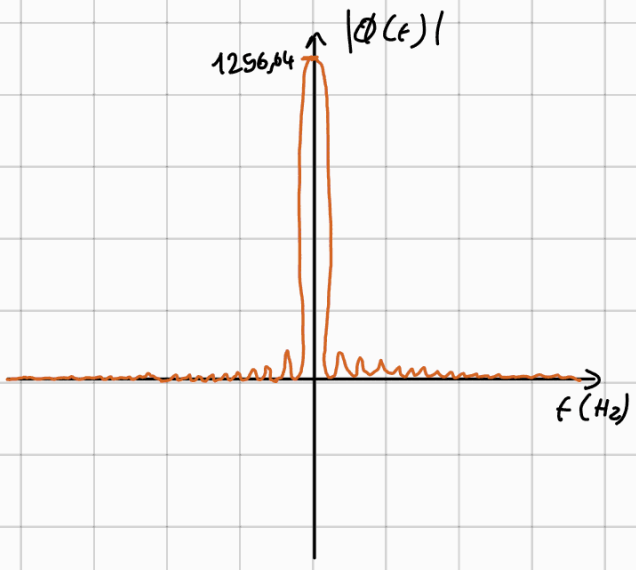
$x(t-t_0) \xleftrightarrow{F} X(f)\,e^{-j2\pi f t_0}$

$$\Lambda\left(\frac{t}{2}\right) \xleftrightarrow{F} 2\,\text{sinc}^2(2f)$$

$$\Lambda\left(\frac{t-2}{2}\right) \xleftrightarrow{F} 2\,\text{sinc}^2(2f)\,e^{-j2\pi f 2}$$

$$\phi(F) = 200\pi \cdot 2\,\text{sinc}^2(2f)\,e^{-j2\pi f 2}$$
$$= 400\pi\,\text{sinc}^2(2f)\,e^{-j2\pi f 2}$$
$$\approx 1256{,}64\,\text{sinc}^2(2f)\,e^{-j2\pi f 2}$$

$$|\phi(f)| = 1256{,}64\,\text{sinc}^2(2f)$$

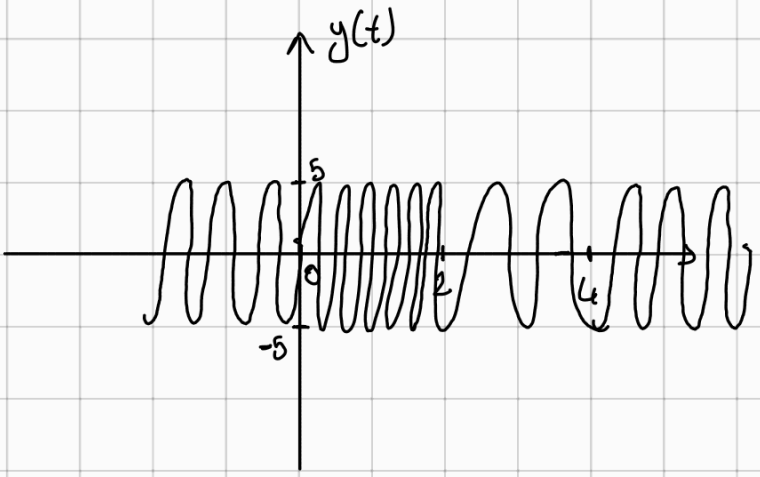


3) $y(t) = 5\cos(500\pi t + \phi(t)) \Rightarrow$

$$y(t) = \begin{cases} 5\cos(500\pi t + 100\pi t) = 5\cos(2\pi 300 t) & 0<t<2 \\ 5\cos(500\pi t + 100\pi(4-t)) = 5\cos(2\pi 200 t + 400\pi) & 2<t<4 \\ \phi(t)=0 \Rightarrow y(t) = 5\cos(2\pi 250 t) & \text{other} \end{cases}$$

(frequency 350; frequency of 200; frequency of 250)

So it can be said that $Y(f)$ will have dominant components at these frequencies.

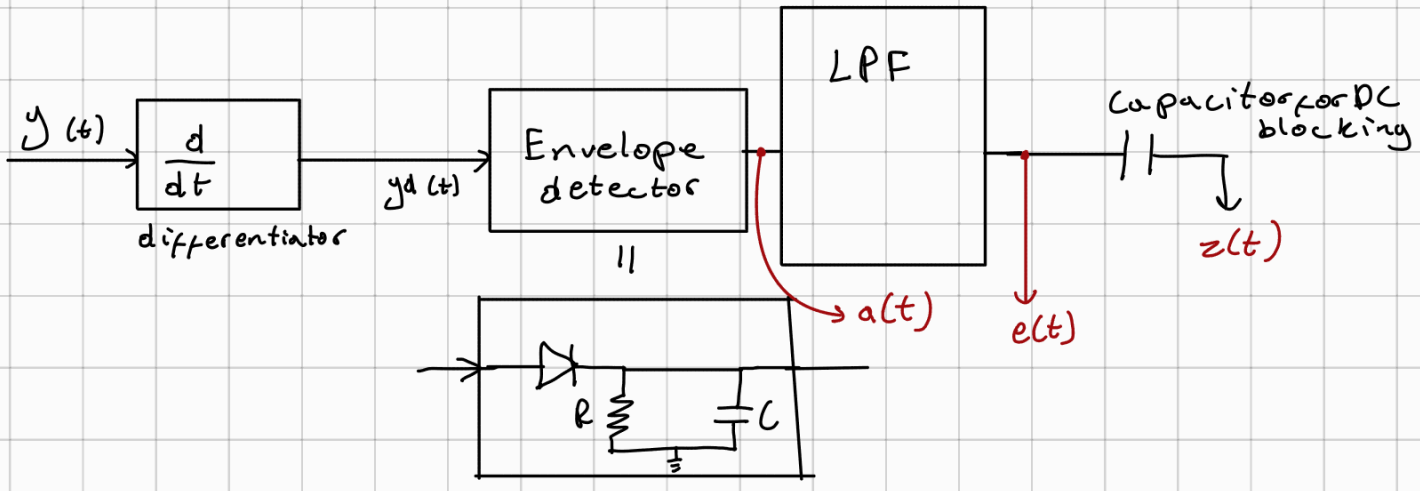


$$\Delta f = k_f m_p = 50 \cdot 1 = 50$$

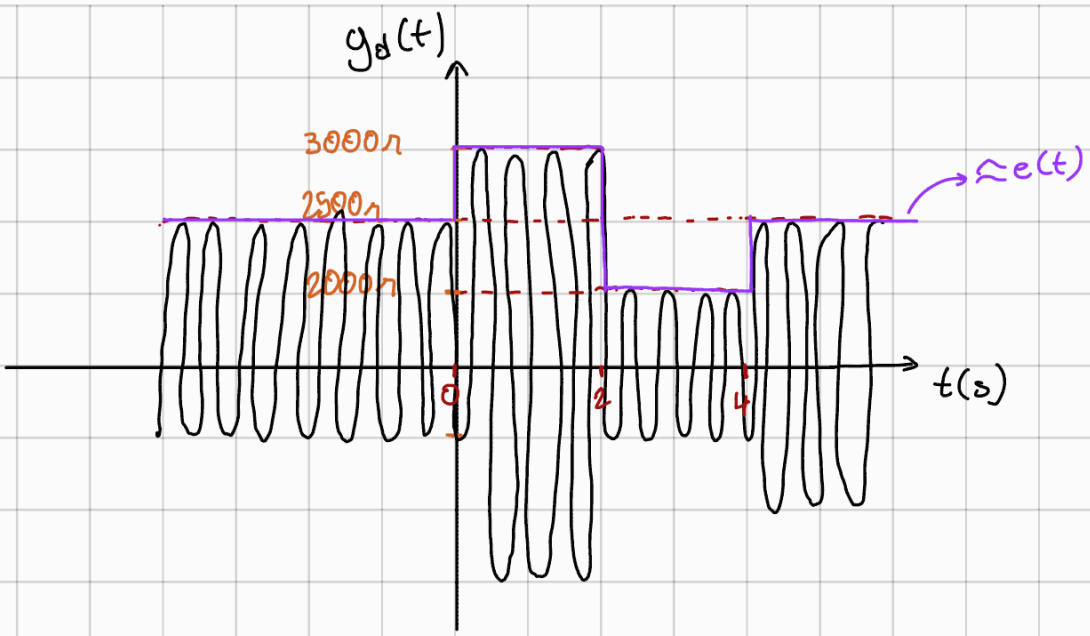
$$\beta = \frac{\Delta f}{B} = \frac{50}{B}$$

$$B_G = 2B(\beta + 1)$$
$$= 2(50 + 2B)$$
$$= (100 + 2B)\ Hz \ //$$

4)



$$y_d(t) = \frac{dy(t)}{dt} = \begin{cases} \frac{d}{dt} 5\cos(2\pi 300 t) & = -3000\pi \sin(2\pi 300 t) & 0<t<2 \\ \frac{d}{dt} 5\cos(2\pi 200 t + 400\pi) & = -2000\pi \sin(2\pi 200 t + 400\pi) & 2<t<4 \\ \frac{d}{dt} 5\cos(2\pi 250 t) & = -2500\pi \sin(2\pi 250 t) & \text{other} \end{cases}$$


$g_d(t)$
3000л
2500л
2000л
≃e(t)
0
2
4
t(s)